RP4 Tests for Ions – Knowledge Organiser

AQA A-level Chemistry • Set 1: Cations and Carbonate **Paper 1**

#	Question	Answer
1	Result for Mg^{2+} with NaOH(aq)	White precipitate
2	Ionic equation for Mg^{2+} with NaOH(aq)	$Mg^{2+} + 2OH^- \rightarrow Mg(OH)_2$
3	Reagent to test for Ba^{2+}	Dilute H_2SO_4
4	Result for Ba^{2+} with dilute H_2SO_4	White precipitate
5	Test for NH_4^+	Warm with NaOH(aq); damp red litmus turns blue
6	Ionic equation for the NH_4^+ test	$NH_4^+ + OH^- \rightarrow NH_3 + H_2O$
7	Test for OH^-	Red litmus turns blue
8	Test for CO_3^{2-}	Add dilute acid; gas turns limewater cloudy
9	Ionic equation for carbonate with acid	$CO_3^{2-} + 2H^+ \rightarrow CO_2 + H_2O$
10	Equation for limewater turning cloudy	$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

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AQA A-level Chemistry • Set 2: Sulfate and Halides **Paper 1**

#	Question	Answer
1	Test for SO_4^{2-}	Acidified $BaCl_2$; white precipitate
2	Ionic equation for the sulfate test	$Ba^{2+} + SO_4^{2-} \rightarrow BaSO_4$
3	Acid used to acidify $BaCl_2$	Dilute HCl
4	Why $BaCl_2$ is acidified	Removes CO_3^{2-} so white $BaCO_3$ cannot form (false positive)
5	Ionic equation for the false positive	$Ba^{2+} + CO_3^{2-} \rightarrow BaCO_3$
6	Test for halide ions	Acidified $AgNO_3$
7	Why $AgNO_3$ is acidified with HNO_3	Removes CO_3^{2-} so Ag_2CO_3 cannot form
8	Ionic equation for the halide test	$Ag^+ + X^- \rightarrow AgX$
9	Order of anion tests	Carbonate, sulfate, halide
10	Why that order is used	Carbonate and sulfate also form precipitates